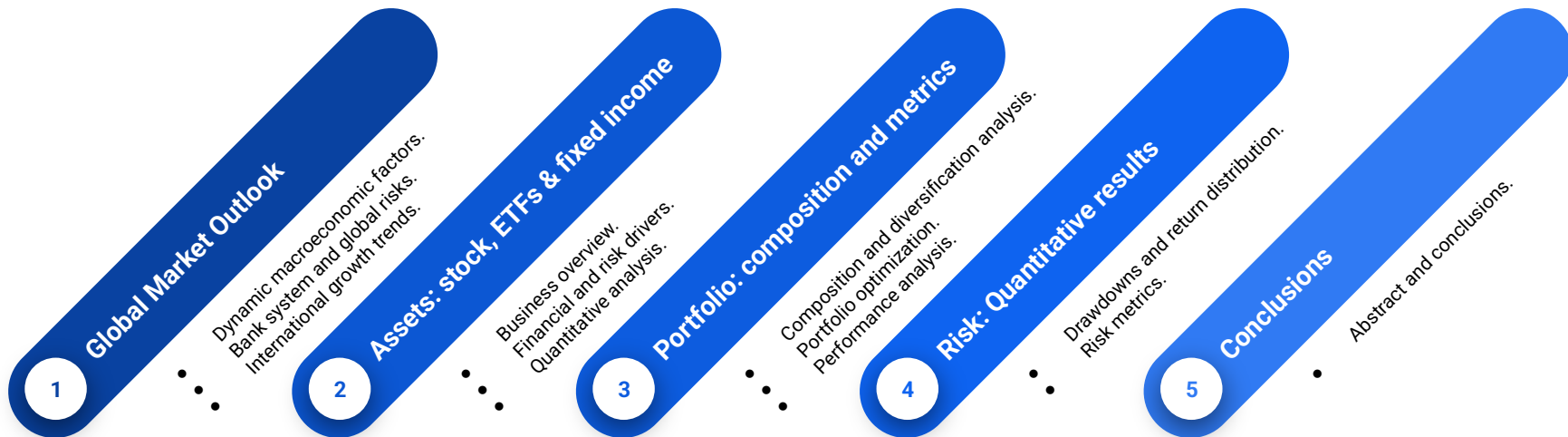




Global Macro-Financial Outlook and US Strategic Drivers for Portfolio Construction

Authors: Caballero, Carlos; Cabrera, Juan & Torre, Karime. Report for UNMSM (2025).

Structure of presentation



Index

Global Market

Assets

Portfolio

Risk

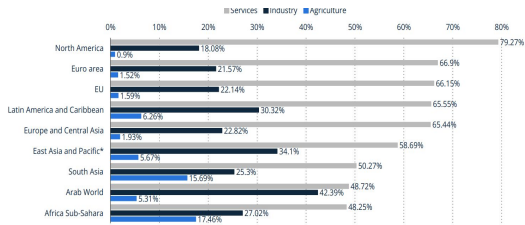
Conclusion

Objectives

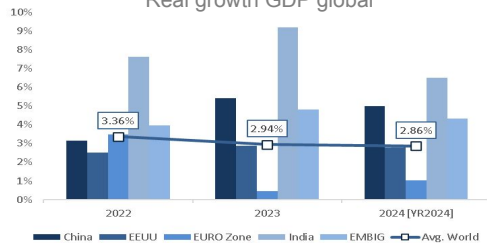
1. Assess key global and regional macroeconomic drivers affecting capital markets.
2. Identify systemic risks (inflation, rates, geopolitics) and their potential market impact.
3. Link macro conditions to investment opportunities and portfolio positioning.

Global Macroeconomic Outlook

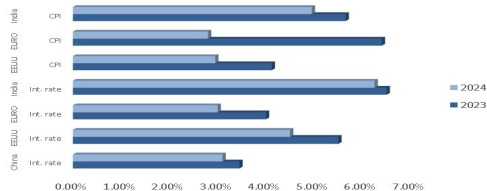
Economic sectors in the GDP of global regions



Real growth GDP global



CPI & Interest rate



Global Economic Indicator

- Global consumer: Global retail sales growth expected around 3% Y/Y (vs 3.5% prepandemic).
- Industrial production growth: Global output growth 2% Y/Y, weak in Asia/Europe.
- Unemployment rate in world: 4.9% (avg.), advanced economies stable but uneven.
- Economic sectors: Services for 60% of global GDP, industry/construction around 25%.

Global Economic Growth Trends

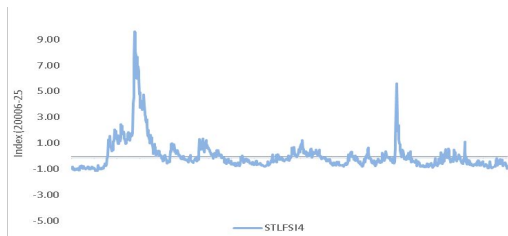
- China slowdown due to real estate stress and weaker external demand.
- Emerging markets remain key contributors, led by India and parts of Asia.
- Global GDP growth moderating to 3%, driven by slower U.S. and Eurozone expansion.
- Growth divergence stems from advanced economies' heavy reliance on the Service sector (71%-79% of GDP), contrasting with other regions' higher dependence on Industry/Agriculture.

Inflation dynamics and monetary policy

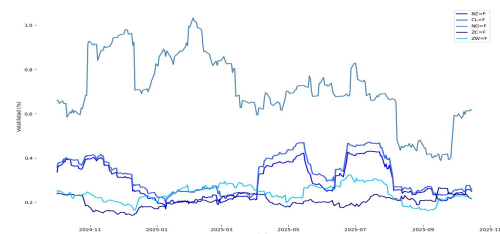
- Inflation cooling but still above central bank targets in major economies.
- Fed and ECB maintaining restrictive policies, delaying rate cuts.
- Sticky services inflation keeps rates “higher for longer.”
- Divergence in monetary policy: developed markets tight, some emerging markets easing.

Global Macroeconomic Outlook

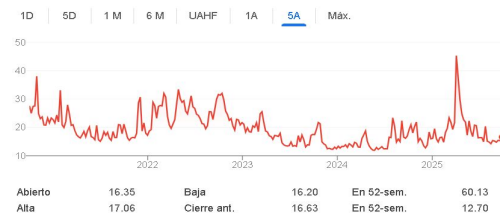
St. Louis Fed Financial Stress Index



Rolling-over of commodities (2024-25)



Chicago Board Options Ex. Market Volatility Index



Financial system's stress

- STLFIS: Near its neutral level, signaling neither expansion nor elevated tension.
- US Fin. System Leverage Index: Below pre-2008 levels, high yields corporates 6× Debt/EBITDA.
- FED: Effective federal funds rate around 4.33% (current).
- Diverging conditions: Advanced economies face slower growth (3%), while emerging markets show stronger growth (5%) but with higher inflation risks.

Geopolitical and energy-related risk

- Ongoing Russia-Ukraine conflict sustaining volatility in energy and grains.
- Rising U.S.–China tensions over trade, tech, and Taiwan.
- Oil price volatility driven by OPEC+ cuts and supply constraints.
- Middle East risks could disrupt global energy flows.

International Capital Markets

- VIX: 16.6 points in the latest close.
- S&P500: Trading near record highs, supported mainly by technology sectors.
- USDX: Strong dollar, appreciating against many EM currencies (+4% over the last 12 months).
- EMBIG: EM spreads remain compressed around 400 bps over Treasuries (country-specific variations).

Tabla 1: Desempeño por clase de activo

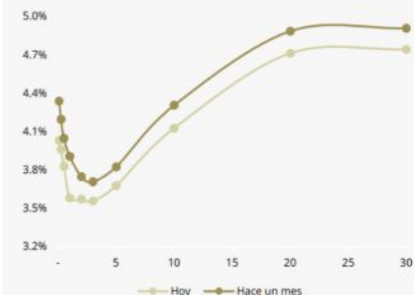
Renta Variable	19-Sep	7D	30D	12M	2025*
RV Global	536.7	0.7%	3.6%	18.3%	17.4%
S&P 500	6,640.0	0.8%	3.8%	16.2%	12.4%
S&P 500 Equal Weight	10,358.4	0.3%	1.6%	7.8%	9.0%
Magnificent 7	32,016.6	3.1%	10.7%	40.7%	17.5%
RV Europa	554.1	-0.1%	-0.9%	6.2%	9.8%
Hang Seng	26,545.1	0.6%	5.5%	47.4%	32.5%
Renta Fija					
RF Global	501.0	0.1%	1.3%	2.7%	8.0%
RF US Treasury	2,413.7	-0.2%	1.3%	1.8%	5.3%
RF US Corporativo	3,520.6	-0.1%	1.8%	3.5%	6.8%
US Treasury 10 Años (%)	4.14	0.07	-0.15	0.42	-0.40
Commodities					
Petróleo (\$/Barril)	66.7	-0.4%	-0.2%	-10.9%	-10.3%
Oro Spot (\$/Onza)	3,672.6	0.8%	9.7%	42.0%	40.9%
Cobre (\$/Libra)	9,940.0	-1.3%	2.3%	4.5%	11.6%
Monedas					
Índice Dólar (\$)	97.59	0.0%	-0.6%	-3.0%	-9.7%
Euro/\$	1.17	0.1%	0.8%	5.2%	12.9%
Renminbi/\$	7.12	-0.1%	-0.8%	0.7%	-2.5%
Sol/\$	3.49	0.1%	-0.8%	-6.8%	-7.1%

(*) Acumulado hasta la fecha.

7D: Últimos 7 días. 30D: Últimos 30 días. 12M: Últimos 12 meses.

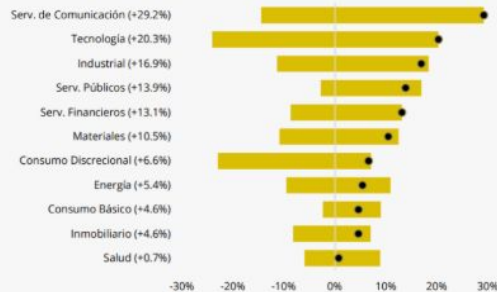
Fuente: Bloomberg. Elaboración: Propia

Gráfico 6: Curva de Rendimientos de Bonos del Tesoro



Fuente: Bloomberg. Elaboración: Propia

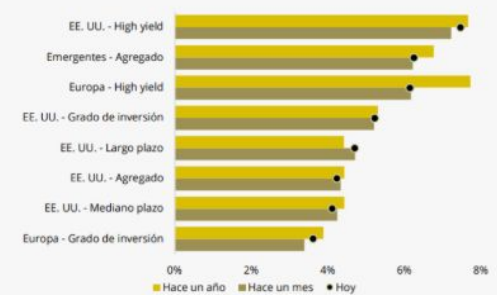
Gráfico 5: Desempeño acumulado por sector - EE. UU.



Sectores medidos a través de los índices de retorno total del S&P 500

Fuente: Bloomberg. Elaboración: Propia

Gráfico 7: Rendimiento de Bonos Globales



Fuente: Bloomberg. Elaboración: Propia

Domestic economic performance

- GDP growth remains resilient but slowing, supported by consumer spending and services.
- Labor market strong, though showing early signs of cooling (wage growth moderating).
- Corporate earnings growth uneven: tech outperforms, while cyclical sectors lag.

Key macro risks

- Sticky inflation delaying Fed rate cuts, raising recession probability.
- Fiscal risks: rising U.S. government debt and deficits.
- Credit risk: stress in commercial real estate and regional banks.
- Political uncertainty: 2024 election cycle could raise market volatility.

US Government Funds

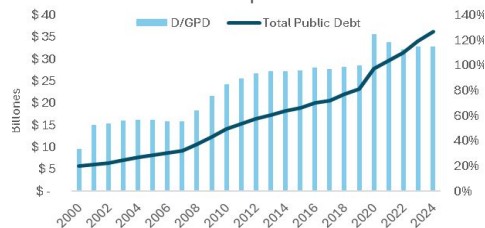
- Deficit: Persistent fiscal gap, higher Treasuries issuance.
- Spending: Defense and infrastructure push demand, risk of crowding-out.
- Debt: Interest burden rising, >15% of budget.
- Sustainability: Fiscal debates affect yields and risk.

EEUU context

Federal Funds Balance



Public debt as a percent of GDP



US Economic Calendar

Hora	Div.	Importancia	Evento	Actual	Previsión	Anterior
Lunes, 29 de septiembre de 2025						
13:30	USD	☆☆☆	Comparecencia de Waller, de la Fed			
16:00	USD	☆☆☆	Venta de viviendas pendientes (Mensual) (Ago)	4.0%	0.2%	-0.3%
18:15	USD	☆☆☆	Declaraciones de Trump, presidente de EEUU			
19:30	USD	☆☆☆	Declaraciones de Williams, miembro del FOMC			

Federal Reserve Assets

- QT: The Fed continues Quantitative Tightening, reducing Treasuries and MBS holdings.
- Liquidity: Asset reduction pressures LT yields upward and lowers structural liquidity.
- Reserves: Bank reserves near the optimal threshold raise the risk of repo market stress.
- Volatility: QT contributes to greater volatility at the long end of the yield curve.

Exchange rate movements and capital flows

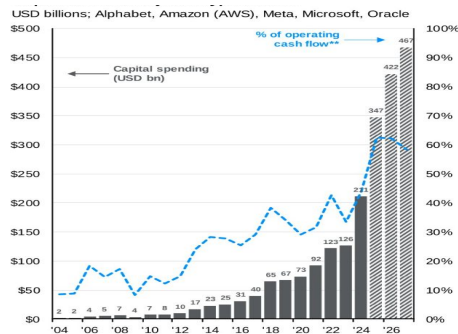
- The U.S. dollar remains strong due to “higher-for-longer” Fed policy.
- Dollar strength pressures emerging markets via capital outflows.
- U.S. Treasuries continue to attract global safe-haven demand.
- Quantitative Tightening (QT) continuing to reduce Treasuries and MBS holdings, pressuring long-term yields upward and lowering structural liquidity.

US Economic Calendar

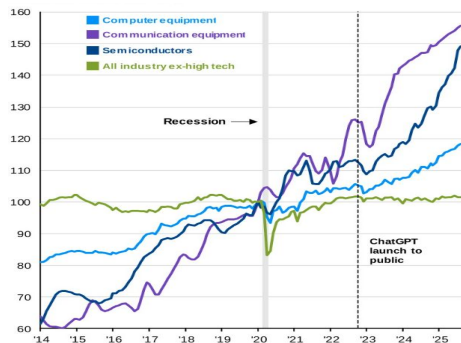
- ISM Services PMI: 50.0 (actual) vs 51.5 (est.) vs 52.0 (prev). Stagnation in services.
- ADP employment: 32K (actual) vs 50K (est.) vs 30K (prev). Negative labor surprise.
- Unemployment rate: 4.3% (actual) vs 4.3% (est./prev). Stable labor market.
- Headline CPI (Y/Y): 2.9% (actual) vs 2.8% (est.) vs 2.7% (prev). Slight uptick in inflation.
- Core CPI (Y/Y): 3.1% (actual) vs 3.0% (est.) vs 3.1% (prev). Sticky underlying inflation.
- GDP growth 25Q2: 3.8% (actual) vs 3.5% (est.) vs -0.6% (prev Q1). Strong rebound.
- Fed Funds rate: 4.33% (actual) vs 4.30% (est.) vs 4.33% (prev). Markets expect mild cuts ahead.

EEUU context

Capex from the major AI hyperscalers



US industrial production of high tech industries



Domestic Investment Trends

- Fixed Investment: Capital spending up 4.2% Y/Y, led by equipment and IP products.
- Private Investment: Solid rebound in nonresidential structures (+8%), especially manufacturing.
- Construction: Manufacturing construction surging >60% YoY from CHIPS & IRA incentives.
- Corporate Capex: Tech and energy firms increasing capex for automation and digitalization.

AI and Innovation

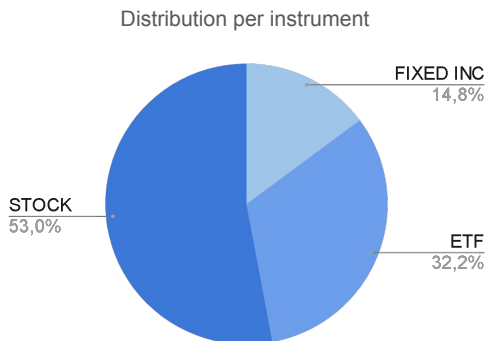
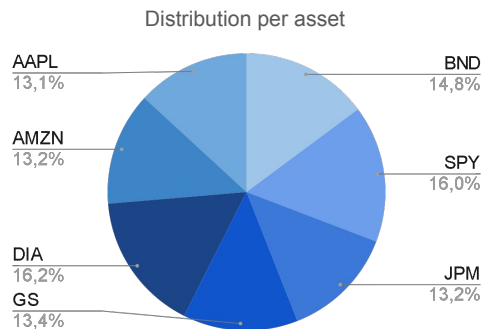
- AI R&D Investment: Over \$200B; concentration in Big Tech, semiconductors and biotech.
- Semiconductor expansion: Fabrication plant projects in states drive industrial GDP.
- AI Adoption: Productivity and profit margin gains across software, healthcare, and manufacturing.
- Tech Ecosystem: Strong venture capital flows and AI startup formation sustain innovation cycle.

Industrial & Sub-Industry Growth

- Semiconductors: Strongest growth driver; fab construction round 70% Y/Y.
- Clean Energy: Renewables and battery storage accelerating under IRA incentives.
- Aerospace & Defense: Federal contracts up 8%, boosting long-term order books.
- Machinery & Equipment: Moderate recovery tied to automation and infrastructure spending.

1. Demonstrate how the final portfolio composition links to the macroeconomic conditions and investment drivers.
2. Optimize the Risk-Return Profile (efficiency) using CML and Ratio Sharpe.
3. Ensure Tactical Diversification to mitigate systemic volatility and achieve comprehensive exposure to the economic cycle.

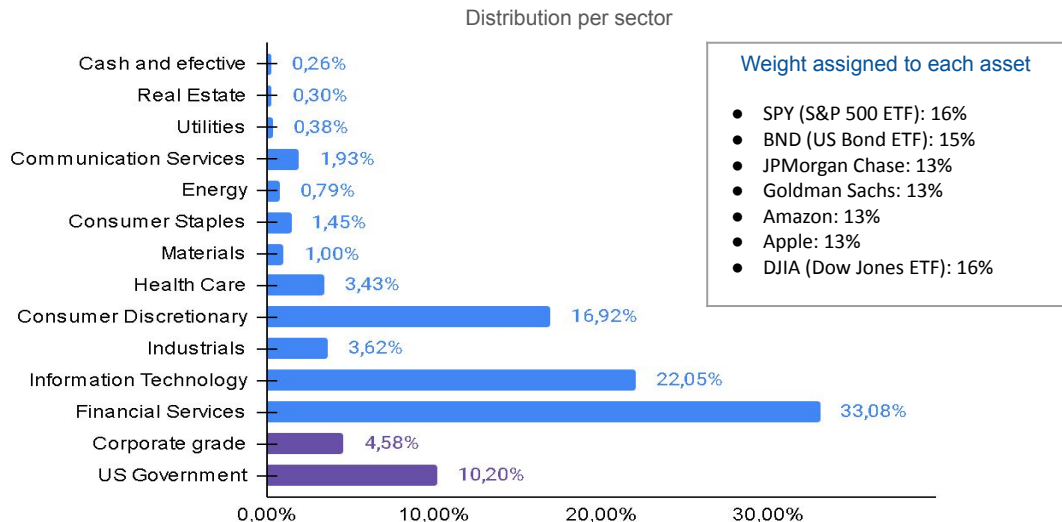
Composition and Diversification Analysis



The 14.8% allocation to Fixed Income (BND) translates into an exposure to the stability of the US Government (10.20%) to offset the risk of Equity

The portfolio seeks to capture the growth of the U.S. market through diversified exposure. It strategically combines high-growth technology stocks (AAPL, AMZN) with defensive assets (JPM, GS, and BND fixed income) to balance returns and stability, mitigating volatility, taking advantage of market dynamism.

The 14.8% allocation to Fixed Income (BND) translates into an exposure to the stability of the US Government (10.20%) to offset the risk of Equity.



Portfolio Optimization and Efficient Frontier

The Markowitz mean-variance model was applied to determine the portfolio's efficient frontier, evaluating two scenarios:

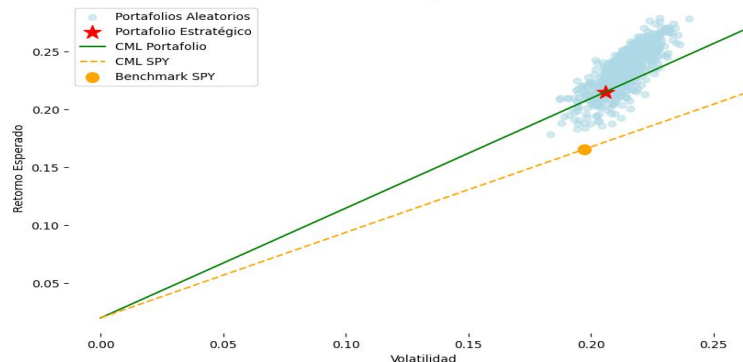
Scenario 1: Weighting limits between 10% and 30% for ETFs, and between 10% and 20% for equities (Sharpe = 1.05 vs 0.75 for SPY).

Scenario 2: An unrestricted version (Sharpe = 1.13) was compared with one with a minimum restriction of +10% per asset (Sharpe = 1.69).

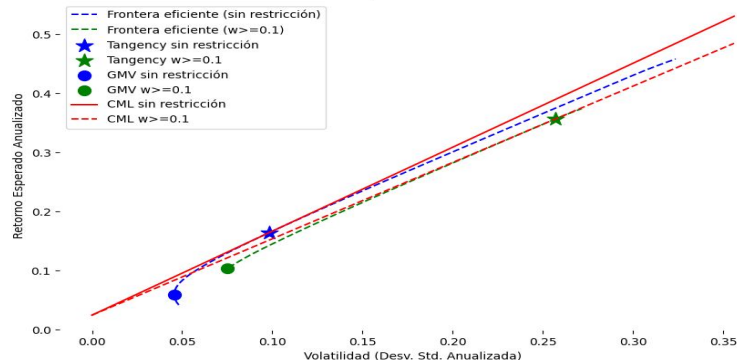
The results show that the efficient portfolio is positioned above the CML, reflecting greater risk-return efficiency compared to the SPY.

The minimum weight restriction improves diversification without significantly increasing volatility, consolidating a structure that combines technological growth and fixed-income stability.

Scenario 1: weighting limits for ETFs and stocks



Scenario 2: unrestricted vs minimum restriction



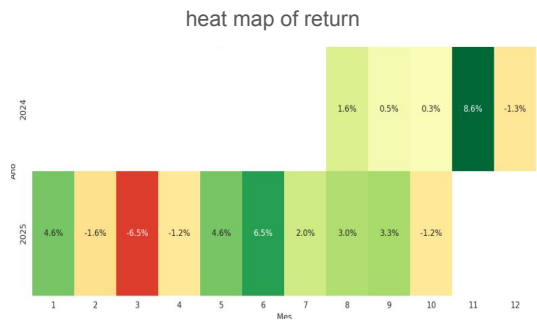
Performance Analysis

The portfolio's performance is due to:

Risk-Adjusted Efficiency: The Sharpe Ratio remains above 1, confirming a risk-adjusted efficiency above the market average.

Volatility Control: The combination of technology, financial, and bond assets has allowed for lower volatility than the benchmark index, thus achieving a favorable risk-return ratio.

Structural Validation: Lower volatility is a direct result of diversification between growth sectors (technology) and defensive assets (financials and bonds).



Sustained Growth: The portfolio has shown sustained growth, achieving a cumulative return of 5% as of August 20, 2025.

Benchmark Outperforming: It has outperformed the SPY benchmark (S&P 500 ETF) over the past year, demonstrating efficient management and a defensive profile.

Return Drivers: Peak returns were concentrated in November 2024 and June 2025, driven by the technological recovery (Apple and Amazon) and the stability of the financial sector.



1. Measure and Control Volatility and Potential Loss.
2. Quantify the portfolio's annualized volatility and perform backtesting for tail risk management.
3. Demonstrate that the diversification strategy has provided resilience to market shocks and kept risk stable over the period.

Distribution Risk: Skewness and Resilience

Stable Volatility: Annualized volatility remained stable throughout the analysis period, even in the face of episodes of high global uncertainty.

Proven Resilience: The portfolio demonstrates resilience to market shocks. This resilience is supported by its diversification across stocks, ETFs, and bonds.

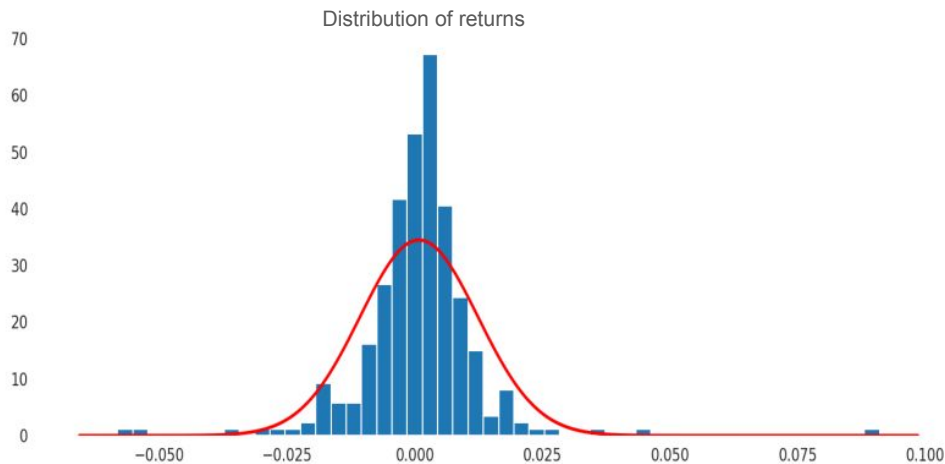
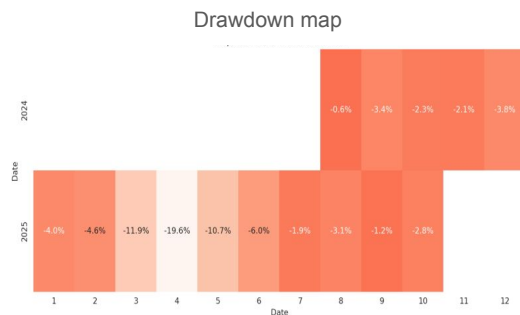
Strategic Balance: Overall, the results confirm a good balance between profitability and stability, which is consistent with its long-term strategy.

Throughout the analyzed period, annualized volatility remained stable, even in the face of episodes of high global uncertainty.

The portfolio exhibits positive skewness in returns, indicating a greater likelihood of moderate gains than extreme losses.

The higher-than-normal kurtosis suggests thicker tails, reflecting exposure to low-frequency but high-impact events, albeit with marginal weight.

Results: Distribution of daily returns (*Sample Days: 289*). *Skewness: 0.8794, Kurtosis: 16.3297, Jarque-Bera Stat: 3248.26, p-value: 0.0000.*



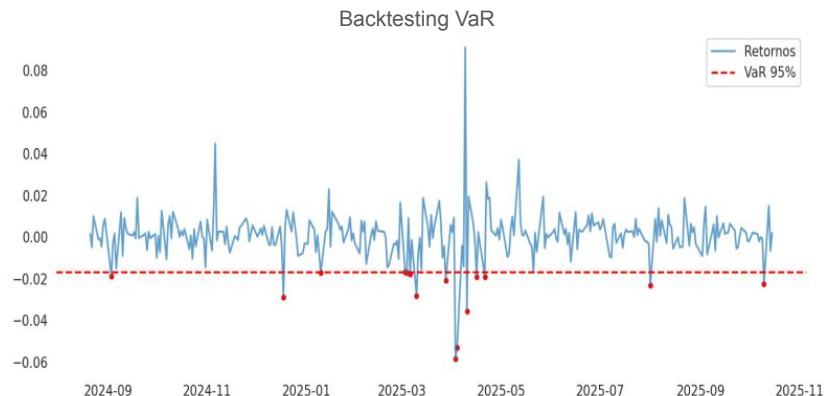
Quantitative Metrics: VaR and Volatility Control

Quantitative risk indicators confirm stability and loss control.

Annualized Volatility is shown to range from 0.1900 to 0.1930. It should be noted that this stability and slight decline (from 0.1930 to 0.1905) is direct evidence of the effectiveness of diversification.

Value at Risk (VaR 95%) demonstrated successful tail risk management for most of the period. Five violations (defaults) were recorded in March-April 2025, an acceptable number in backtesting.

Conclusion: The graph of daily returns versus 95% VaR shows that only in a few cases did negative returns exceed the VaR threshold. This indicates that the portfolio successfully managed tail risk for most of the analysis period.



Conclusion

The portfolio successfully implements a top-down strategy that capitalizes on key US market trends, demonstrating risk-adjusted efficiency above the market average.

- **Performance and Efficiency: Benchmark Outperforming:** The portfolio consistently outperformed the SPY benchmark (S&P 500 ETF), maintains a Sharpe Ratio above 1. Profitability was primarily driven by technological dynamism and the stability of the financial sector.
- **Risk Management and Stability: Diversification** resulted in lower volatility than the benchmark index, achieving a favorable balance between risk and return. The portfolio demonstrates resilience to market shocks, maintaining a positive skewness in returns.
- Also, the high concentration in Financial Services and Technology is offset by the inclusion of defensive assets such as U.S. government debt, which is vital to mitigate the latent risks of inflation and the Fed's Quantitative Tightening (QT) policies.